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ARTICLE OF JEWELRY WITH MAGNETIC CLASP ASSEMBLY

Abstract

A magnetic clasp assembly includes a first member and a second member each couplable to an end of a bracelet. The first member may be readily secured or removed from the bracelet by actuating a first lever. Likewise, the second member may be readily secured or removed from the bracelet by actuating a second lever. The second member may be coupled to the first member via a second protrusion on the second member being inserted into a first receiving slot of the first member, and a first protrusion couplable to a second receiving slot of the second member. Magnets may be used to ensure the first and second member remain coupled. By actuating either the first or second lever, attachment and removal of slider charms and other ornamental accessories from the bracelet is facilitated. Such slider charms may feature designs representing a user's life achievements and interests.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. Non-Provisional application Ser. No. 18/352,139, filed on Jul. 13, 2023, which claimed the benefit of U.S. Provisional Application Ser. No. 63/377,786, filed on Sep. 30, 2022 and U.S. Provisional Application Ser. No. 63/368,442, filed on Jul. 14, 2022, all of which are incorporated herein by reference.

TECHNICAL FIELD
[0002] The present disclosure relates to jewelry fasteners. More particularly, this disclosure relates to an article of jewelry with a magnetic clasp assembly.
BACKGROUND
[0003] A jewelry clasp, or jewelry fastener, is the mechanical mechanism that allows a necklace or bracelet to easily be put on and taken off without causing any damage to the jewelry. Some fasteners are meant to be hidden, while others are a key element to the overall design. Common variants of jewelry fasteners include ball, slide, swivel, push button, fishhook, lobster, and buckle clasps. Such closure mechanisms, however, can be difficult to quickly attach and remove, uncomfortable to wear, and distract aesthetically from other elements of the jewelry.
[0004] A charm bracelet, in particular, is a type of bracelet which carries personal jeweled ornaments or “charms,” such as decorative pendants and trinkets. The decorative charms usually carry personal or sentimental attachment for the user. In some cases, a charm bracelet can further be used to display and celebrate the user's accomplishments and significant life events.
[0005] Interchanging which charms are displayed on the charm bracelet or other article of jewelry may require the user to frequently open and close the jewelry fastener and slide the charms both over the jewelry fastener and along the band of the bracelet. This process is complicated if the jewelry fastener is too bulky and thus unable to be readily threaded through the charms. Moreover, the jewelry fastener is essential to creating a comfortable fit around the wrist or neck and most jewelry fasteners do not enable the user to adjust the length of the band of the bracelet without removing links or substituting the standard band of the bracelet for a smaller or larger size.
[0006] Accordingly, there is a need for an article of jewelry, in particular a charm bracelet, with a detachable jewelry fastener that easily facilitates the interchange of new charms and accessories, and which permits adjustment of the bracelet's sizing. The present disclosure solves these and other problems.
SUMMARY OF EXAMPLE EMBODIMENTS
[0007] In some embodiments, an article of jewelry comprises a bracelet and a magnetic clasp assembly detachably couplable to the bracelet. The magnetic clasp assembly further comprises a first member removably couplable to a first end of the bracelet and a second member coupled to a second end of the bracelet. The first member comprises a first engagement area, a lever arm groove, and a lever, wherein the first engagement area comprises a first magnet cavity, a first magnet couplable to the first magnet cavity, a first protrusion, and a first receiving slot. The first member may further comprise a first aperture, a second aperture, a channel between the first and second apertures, and a pin couplable through the channel to the lever, wherein the lever is configured to be actuated in relation to the first member. The second member comprises a second

engagement area couplable to the first engagement area. The second engagement area comprises a second magnet cavity, a second magnet couplable to the second magnet cavity, a second protrusion, and a second receiving slot. In some embodiments, the lever of the article of jewelry with magnetic clasp assembly further comprises a plurality of protrusions opposite a plurality of ridges on the first member. Interaction between the plurality of protrusions and the plurality of ridges enables greater friction between the bracelet and the first member, preventing unwanted withdrawal of the bracelet from the first member. Actuating the lever facilitates removal of the first member, allowing a user to add or remove charms to the bracelet.
[0008] In some embodiments, a method of using an article of jewelry comprises coupling a first member to a second member of a magnetic clasp assembly by coupling the second protrusion with the first receiving slot, coupling the first protrusion with the second receiving slot, and engaging through proximity the first magnet and the second magnet. A user may readily couple the bracelet to the first member by placing a band, chain, or rope of an article of jewelry into a bracelet cavity of the first member, and actuating the lever from an open configuration to a closed configuration. The first member may be removed from the bracelet to facilitate the addition or removal of charms by actuating the lever in the reverse direction.
[0009] In some embodiments, both the first member and second member of the clasp assembly each comprise a lever configured to allow each member to be removably attachable to a bracelet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0010] FIG. **1** illustrates a top, side perspective view of an article of jewelry with a magnetic clasp assembly;
[0011] FIG. **2** illustrates a left side elevation view of a magnetic clasp assembly;
[0012] FIG. **3** illustrates a top, left side perspective view of a magnetic clasp assembly disassembled;
[0013] FIG. **4** illustrates a top, left side cross-sectional view of a magnetic clasp assembly disassembled;
[0014] FIG. **5** illustrates a top, side perspective view of a magnetic clasp assembly disassembled;
[0015] FIG. **6** illustrates a top, side perspective view of a magnetic clasp assembly assembled;
[0016] FIG. **7** illustrates an exploded view of a magnetic clasp assembly;
[0017] FIG. **8** illustrates a top, side perspective view of a first member;
[0018] FIG. **9** illustrates a left side elevation, sectional view of a first member;
[0019] FIG. **10** illustrates a top, front perspective view of a lever;
[0020] FIG. **11** illustrates a top, front perspective view of a slider charm;
[0021] FIG. **12** illustrates a left, side cross-sectional view of a magnetic clasp assembly;
[0022] FIG. **13** illustrates a left, bottom perspective view of a magnetic clasp assembly disassembled; and
[0023] FIG. **14** illustrates a left, cross-sectional view of a second member of a magnetic clasp assembly.
DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS
[0024] The following descriptions depict only example embodiments and are not to be considered limiting in scope. Any reference herein to “the invention” is not intended to restrict or limit the invention to exact features or steps of any one or more of the exemplary embodiments disclosed in the present specification. References to “one embodiment,” “an embodiment,” “various embodiments,” and the like, may indicate that the embodiment(s) so described may include a particular feature, structure, or characteristic, but not every embodiment necessarily includes the particular feature, structure, or characteristic. Further, repeated use of the phrase “in one

embodiment," or "in an embodiment," do not necessarily refer to the same embodiment, although they may.
[0025] Reference to the drawings is done throughout the disclosure using various numbers. The numbers used are for the convenience of the drafter only and the absence of numbers in an apparent sequence should not be considered limiting and does not imply that additional parts of that particular embodiment exist. Numbering patterns from one embodiment to the other need not imply that each embodiment has similar parts, although it may.
[0026] Accordingly, the particular arrangements disclosed are meant to be illustrative only and not limiting as to the scope of the invention, which is to be given the full breadth of the appended claims and any and all equivalents thereof. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation. Unless otherwise expressly defined herein, such terms are intended to be given their broad, ordinary, and customary meaning not inconsistent with that applicable in the relevant industry and without restriction to any specific embodiment hereinafter described. As used herein, the article "a" is intended to include one or more items. When used herein to join a list of items, the term "or" denotes at least one of the items, but does not exclude a plurality of items of the list. For exemplary methods or processes, the sequence and/or arrangement of steps described herein are illustrative and not restrictive.
[0027] It should be understood that the steps of any such processes or methods are not limited to being carried out in any particular sequence, arrangement, or with any particular graphics or interface. Indeed, the steps of the disclosed processes or methods generally may be carried out in various sequences and arrangements while still falling within the scope of the present invention.
[0028] The term "coupled" may mean that two or more elements are in direct physical contact. However, "coupled" may also mean that two or more elements are not in direct contact with each other, but yet still cooperate or interact with each other.
[0029] The terms "comprising," "including," "having," and the like, as used with respect to embodiments, are synonymous, and are generally intended as "open" terms (e.g., the term "including" should be interpreted as "including, but not limited to," the term "having" should be interpreted as "having at least," the term "includes" should be interpreted as "includes, but is not limited to," etc.).
[0030] As previously discussed, there is a need for an article of jewelry, in particular a charm bracelet, with a detachable magnetic clasp assembly that easily facilitates the exchange of new charms and accessories, and which also permits adjustment of the bracelet's sizing. The present disclosure addresses these and other problems
[0031] As illustrated in FIG. **1**, in some embodiments, an article of jewelry with magnetic clasp assembly **100** comprises a bracelet **102** and a magnetic clasp assembly **104** detachably couplable to the bracelet **102**. The bracelet may comprise a band, chain, rope, or other form factor that is capable of being fed through the magnetic clasp assembly **104** and secured while being worn. The bracelet **102** may alternatively be a necklace, anklet, armband, or related article of jewelry worn on other parts of the body. The magnetic clasp assembly **104** is configured to grip and release material of the bracelet **102** via a lever **114** in order to add or remove slider charms **148**, as well as to adjust a size of the bracelet **102** as desired by the user.
[0032] As illustrated in FIG. **2**, in some embodiments, the magnetic clasp assembly **104** comprises a first member **106** and a second member **108** detachably couplable to the first member **106**. As shown, though without limitation, the first member **106** and the second member **108** may be substantially symmetrical in shape and configured to overlap with one another, thereby forming a complete magnetic clasp assembly **104** that is smooth and rectangular when fastened to the bracelet **102**. It will be appreciated, however, by those skilled in the art, that the first member **106** and the second member **108** of the magnetic clasp assembly **104** may be asymmetrical and couplable together to form other shapes.
[0033] Referring to FIGS. **3-7**, the first member **106** comprises a first engagement area **110**, a lever

arm groove **112** couplable to the lever **114**. The first engagement area **110** comprises a first magnet cavity **116**, a first magnet **118** couplable within the first magnet cavity **116**, a first protrusion **120**, and a first receiving slot **122**. The second member **108** comprises a second engagement area **134** couplable to the first engagement area **110**, and a compartment **152** for receiving an end of the bracelet **102** (which may be secured therein using adhesives, a set screw, or similar securing mechanism). The second engagement area **134** comprises a second magnet cavity **150**, a second magnet **138** couplable within the second magnet cavity **150**, a second protrusion **140**, and a second receiving slot **142**.

[0034] A first surface of the first magnet **118** may be configured to be a north pole and a first surface of the second magnet **138** may be configured to be a south pole or vice versa to permit attraction between the first and second magnets **118**, **138** when placed in close proximity to one another. Interaction between the first and second magnets **118**, **138** facilitates fastening together of the first member **106** with the second member **108** of the magnetic clasp assembly **104** as a whole (e.g., FIG. **2**). Additionally, the first protrusion **120** is couplable to the second receiving slot **142** through interlocking or compression fit and the second protrusion **140** is likewise couplable to the first receiving slot **122** by interlocking or compression fit. These additional means of securement ensure proper alignment of the first member **106** and second member **108**. For example, a user would tilt the second member **108** such that the second protrusion **140** is insertable into the first receiving slot **122**. Once received, the second member may then be pressed downwardly such that the second receiving slot **142** receives the first protrusion **120**. As the second magnet **138** approximates the first magnet **118**, they are attracted to one another, securing the second member **108** to the first member **106**. Accordingly, the user may decouple the first member **106** from the second member **108** only through intentional application of force to break the magnetic field and to remove the second protrusion **140** from the first receiving slot **122** at the correct angle.

[0035] As best seen in FIG. **7**, the lever **114** comprises an arm **124**, a first aperture **126**, a second aperture **128**, a channel **130** between the first and second apertures **126**, **128**, and a pin **132** receivable through the channel **130** that couples the lever **114** within a bracelet cavity **158** of the first member **106** via pin apertures **115**A-B. As appreciated, the pin **132** functions as a hinge for the lever **114**. In some embodiments, as illustrated in FIG. **8**, the lever arm groove **112** may be rounded rather than square, with the lever arm **124** complementary in shape, so as to be comfortable against a user's arm. Moreover, as best seen in FIG. **9**, the first member **106** may have a plurality of ridges **156** within the bracelet cavity **158** to engage with and secure the bracelet **102** between the lever **114** and other ridges, as discussed in more detail later herein. The plurality of ridges **156**, in some embodiments, may be positioned on an internal surface of the first member **106** and comprise, without limitation, narrow raised bands having a square shape that are spaced at even intervals to create friction against the bracelet when positioned within the bracelet cavity **158**.

[0036] Speaking functionally, the lever **114** may be actuated in order to couple the magnetic clasp assembly **104** to the bracelet **102**. When the lever **114** is in an open configuration **144**, as best seen in FIGS. **3-4**, an end of the bracelet **102** or other jewelry such as the band, chain, or rope may be fed into the bracelet cavity **158**, interposed between the top end **160** of the lever **114** and the plurality of ridges **156** of the first member **106**. Next, the lever **114** may be actuated by forcing the lever arm **124** into the lever arm groove **112**. In so doing, a raised end **162** applies a force to the bracelet **102**, pinching the bracelet **102** between the raised end **162** and the plurality of ridges **156**, forming a closed configuration **146**, as best seen in FIGS. **1-2**.

[0037] In some embodiments, as shown in FIG. **10**, a lever **214** may comprise a lever arm **224**, a top end **260**, and a raised end **262** comprising a plurality of protrusions **264**. The plurality of protrusions **264** apply additional pressure to the bracelet **102** against the plurality of ridges **156** such that when the bracelet **102** is interposed between the plurality of protrusions **264** and the plurality of ridges **156**, the bracelet is prevented from being withdrawn.

[0038] When the first and second members **106**, **108** are coupled together with the lever **114** (or

214) in the closed configuration **146** (e.g., FIG. **6**), the arm **124** of the lever **114** may be secured within the groove **112**, providing a comfortable fit along the wrist or neck of the user. The arm **124** of the lever **114** may further have a slight curvature on at least one side corresponding to the curvature of the wrist or neck of the user to further provide that comfortable fit.

[0039] It will be appreciated that the magnetic clasp assembly **104** and the slider charms **148** (FIGS. **1** and **11**) may be laser engraved or otherwise feature designs representing the user's life achievements and interests. The slider charms **148** may be freely interchanged with ease according to the user's daily preferences.

[0040] It will be appreciated that the first member **106** of the magnetic clasp assembly **104** may be completely removable from the bracelet **102**, permitting the user to freely add or remove slider charms **148** to the bracelet **102**. The slider charms **148** are retained on the bracelet **102** via the first and second members **106**, **108**. In other words, an inner circumference of the slider charm **148** is less than an outer circumference of the first and second members **106**, **108** such that the slider charms **148** are not capable of sliding over the first and second members **106**, **108**. Therefore, to add or remove slider charms **148** from the bracelet **102**, a user may remove the first member **106** by actuating the lever **114** to release the tension on the bracelet **102** (e.g., raised end **162** rotates to disengage with the bracelet **102**), and thereby slide the bracelet **102** out of the bracelet cavity **158**. The user may then add or remove slider charms **148** to the bracelet **102** as desired, then feed the bracelet **102** back through into the bracelet cavity **158** (i.e., between the top end **160** of the lever and the ridges **156**) to the desired position (sized to the user's wrist, for example), then actuate the lever **114** again to retain the bracelet **102** between the ridges **156** and the raised end **162** (or protrusions **264**). The user may then assemble the second member **108** to the first member **106** as described earlier herein, where they are secured to one another via the coupling of the first and second magnets **118**, **138**, as well as the coupling of the second protrusion **140** in the first receiving slot **122** and the first protrusion **120** in the second receiving slot **142**. A user may easily remove the bracelet **102** by simply disconnecting the first member **106** from the second member **108**. However, by simply uncoupling the first member **106** from the second member **108**, the slider charms **148** are retained on the bracelet **102** as they cannot slide over either the first or the second member **106**, **108**.

[0041] While the discussion above illustrates a lever **114** on the first member **106**, it will be appreciated that the second member may also comprise a lever. For example, FIGS. **12-14** illustrate a magnetic clasp assembly **200** comprising the first member **106** and a second member **202**. Like the first member **106**, the second member **202** comprises a lever **204** configured to pivot on a pivot pin **206** between an open configuration, wherein the bracelet is removable from the second member **202**, and a closed configuration, wherein the bracelet is not removable from the second member **202**. Similar to the first member **106**, a bracelet cavity **208** is positioned between a raised end **210** of the lever **204** and an upper surface **212** of the second member **202**.

[0042] As shown, the raised end **210** may comprise a plurality of teeth **214**A, or other protrusions, to apply pressure to the bracelet positioned within the bracelet cavity **208**. The upper surface **212** may likewise comprise one or more ridges **216** or other protrusions to likewise aid in creating a non-slip surface, thereby preventing unwanted withdrawal of the bracelet from the bracelet cavity **208** when the lever **204** is in the closed configuration (as shown in FIGS. **12-14**). Additionally, as shown, the lever **114** of the first member **106** may comprise similar features, such as teeth **214**B. However, while both the first and second member **106**, **202** are shown with teeth **214**A-B, it will be appreciated that teeth **214**A-B are not required, and that simply comprising a raised portion (e.g., **162**) as disclosed and shown earlier herein is contemplated.

[0043] The second member **202** may also comprise a lever arm groove **218** that allows a user to easily grab and withdraw the lever **204**. The lever arm groove **218** may also allow the lever **204** to remain flush with the bottom of the first member **106** when in a closed configuration. When the lever **204** is actuated by a user to an open configuration (e.g., lever **204** pivoted about ninety degrees, similar to that shown at **144** in FIG. **4**), the bracelet is no longer clamped between the ridges **216** and the lever **204**, allowing the bracelet to be removed therefrom. As shown, the first member **106** and second member **202** couple together in the same manner as previously disclosed herein with other embodiments, including the features disclosed therein. Additionally, while the lever **204** of the second member **202** is shown with a length smaller than that of the first lever **114**, this configuration is not required, and the levers **114**, **204** may have differing lengths or the same lengths without departing herefrom.

[0044] Therefore, it will be appreciated that a magnetic clasp assembly may comprise a first member **106** with a lever **114** couplable to a second member **108** without a lever, or couplable to a second member **202** with a lever **204**.

[0045] Ease of use permits the user to interchange slider charms **148** more frequently and thereby customize the user's jewelry to display different life events, new achievements, seasonal holidays, awareness and support for significant causes, and other activities. Moreover, the capacity to place the magnetic clasp assembly **104** at any segment of the bracelet **102**, particularly when using a braded rope bracelet **102**, allows the user to achieve a comfortable custom fit that also accommodates the chosen quantity of slider charms **148** added to the bracelet **102**. Accordingly, the article of jewelry with magnetic clasp assembly **100** solves the need for an article of jewelry that easily facilitates the interchange of new slider charms **148** and accessories, and which permits adjustment of the bracelet's **102** sizing around the wrist or neck, overcoming limitations in the prior art.

[0046] It will be appreciated that systems and methods according to certain embodiments of the present disclosure may include, incorporate, or otherwise comprise properties or features (e.g., components, members, elements, parts, and/or portions) described in other embodiments. Accordingly, the various features of certain embodiments can be compatible with, combined with, included in, and/or incorporated into other embodiments of the present disclosure. Thus, disclosure of certain features relative to a specific embodiment of the present disclosure should not be construed as limiting application or inclusion of said features to the specific embodiment unless so stated. Rather, it will be appreciated that other embodiments can also include said features, members, elements, parts, and/or portions without necessarily departing from the scope of the present disclosure.

[0047] Moreover, unless a feature is described as requiring another feature in combination therewith, any feature herein may be combined with any other feature of a same or different embodiment disclosed herein. Furthermore, various well-known aspects of illustrative systems, methods, apparatus, and the like are not described herein in particular detail in order to avoid obscuring aspects of the example embodiments. Such aspects are, however, also contemplated herein.

[0048] Exemplary embodiments are described above. No element, act, or instruction used in this description should be construed as important, necessary, critical, or essential unless explicitly described as such. Although only a few of the exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in these exemplary embodiments without materially departing from the novel teachings and advantages herein. Accordingly, all such modifications are intended to be included within the scope of this invention.

Claims

1. An article of jewelry, comprising: a bracelet; a slider charm configured to be slidable on the bracelet; and a magnetic clasp assembly, comprising: a first member configured to be removably attachable to the bracelet, the first member, comprising: a first engagement area, and a lever, wherein the lever is selectively actuatable between a closed configuration wherein the lever prevents withdrawal of the bracelet from the first member, and an open configuration wherein the lever releases the bracelet from the first member; and a second member configured to be removably attachable to the bracelet, the second member, comprising: a second engagement area complementary in shape to the first engagement area of the first member, and a second lever, wherein the second lever is selectively actuatable between a closed configuration wherein the second lever prevents withdrawal of the bracelet from the second member, and an open configuration wherein the second lever releases the bracelet from the second member.

2. An article of jewelry, comprising: a bracelet; a slider charm configured to be slidable on the bracelet; and a magnetic clasp assembly, comprising: a first member configured to be removably attachable to the bracelet, the first member, comprising: a first engagement area, a lever comprising a lever arm, a top end, and a raised end, the lever arm configured to selectively abut a lever arm groove via pivoting on a pin, a bracelet cavity comprising a plurality of ridges, the plurality of ridges extending toward the top end of the lever in the bracelet cavity, wherein in a first position, the lever arm is substantially perpendicular to the first engagement area and the top end is proximal to the ridges, allowing the bracelet to be inserted or removed into the bracelet cavity, and in a second position, the lever arm abuts the lever arm groove and the raised end is proximal to the ridges, thereby interposing the bracelet between the raised end and plurality of ridges, thereby preventing withdrawal of the bracelet; and a second member configured to be removably attachable to the bracelet, the second member, comprising: a second engagement area complementary in shape to the first engagement area of the first member, a second lever comprising a lever arm, a top end, and a raised end, the lever arm configured to selectively abut a lever arm groove via pivoting on a second pin, a bracelet cavity comprising a plurality of ridges, the plurality of ridges extending toward the top end of the lever in the bracelet cavity, wherein in a first position, the lever arm is substantially perpendicular to the first engagement area and the top end is proximal to the ridges, allowing the bracelet to be inserted or removed into the bracelet cavity, and in a second position, the lever arm abuts the lever arm groove and the raised end is proximal to the ridges, thereby interposing the bracelet between the raised end and plurality of ridges, thereby preventing withdrawal of the bracelet.

3. The article of jewelry of claim 2, wherein an inner circumference of the slider charm is less than an outer circumference of the first member and the second member.

4. The article of jewelry of claim 2, wherein the bracelet comprises a band, chain, or rope.

5. A method of using an article of jewelry with magnetic clasp assembly, the method comprising: sliding a charm on a bracelet; actuating a lever arm of a first member so that the lever arm is substantially perpendicular to a first engagement area of the first member; coupling the first member to a first end of the bracelet by inserting the first end into a bracelet cavity; actuating the lever arm of the first member so that the first end of the bracelet is interposed between a raised end of a lever and a plurality of ridges to secure the bracelet in the bracelet cavity, the lever arm abutting a lever arm groove; coupling a second protrusion of a second member to a first receiving slot of the first member; coupling a first protrusion of the first member to a second receiving slot of the second member, which simultaneously couples a first magnet to a second magnet; and actuating the lever between an open configuration and a closed configuration to add or remove charms from the bracelet.
